TEXORA · LECTURE SHEET

Pressing and Finishing

Heat, moisture, pressure and vacuum applied as a controlled process: under-pressing and top pressing, fibre temperature limits, the finishing-section equipment, pressing faults, and end-of-line finishing.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 8 · LEVEL 3 TERM 1

1. Why pressing is manufacturing, not cosmetics

Pressing (ironing/finishing) applies **heat, moisture, pressure and vacuum/cooling** in controlled combination to set a garment's final shape: removing creases, sharpening designed creases, moulding three-dimensional form (a jacket shoulder, a trouser crease) and presenting the product the customer actually judges. It is done twice in a garment's life in the factory: **under-pressing (in-process)** — pressing seams open, fusing, shaping parts during making, without which sewing quality cannot be held — and **top pressing (final)** after sewing and washing checks, before folding and packing (Topic 9).

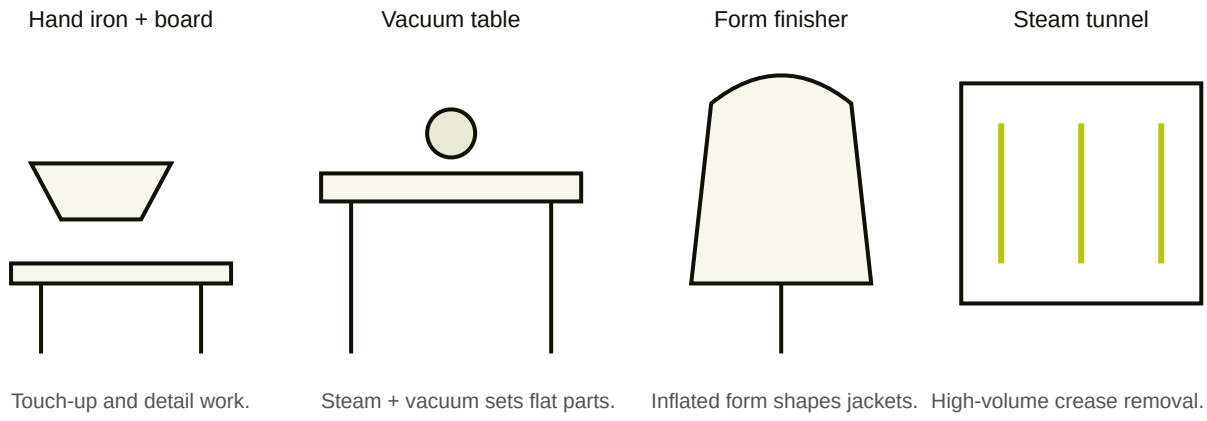


Figure 1 — The four pressing stations of a finishing section, from hand work to the steam tunnel.

2. The four variables and the fibre rules

VARIABLE	ROLE
Heat	Softens fibres so they take a new set; limits are fibre-specific (below).
Moisture (steam)	Plasticises fibres (especially wool and cotton) so shaping is fast and durable; also removes creases by relaxing stored stresses.

Pressure	Flattens seams, forms creases and compresses the structure while it sets.
Vacuum / cooling	Removes moisture and cools the part under shape, fixing the set and preventing re-creasing during handling.

FABRIC	IRON SOLE GUIDE	NOTES
Cotton / linen	High (~200 °C)	Needs steam for creases; scorches only near the top of the range.
Wool	Medium-high (~160 °C) with steam and a press cloth	Direct dry contact glazes (shines) the surface; steam-and-vacuum is the wool method.
Polyester	Medium (~150 °C)	Heat-sets permanently — a wrong crease is also permanent; melts/glazes if overheated.
Nylon	Low-medium (~130 °C)	Softens and sticks at modest temperatures; test first.
Acrylic / elastane blends	Low (~110–120 °C)	Elastane degrades with heat: minimum temperature, no prolonged contact.

3. Equipment of the finishing section

- **Steam iron and vacuum table** — the universal station: steam from a boiler through the iron, suction through the table board; shaped bucks (sleeve boards, ham/tailor's cushions) support curved parts.
- **Form finisher (steam dolly)** — the garment is dressed on an inflatable form; steam and hot air relax creases from inside; jackets, dresses, blouses.
- **Trouser press / legger and topper** — the legger presses both legs' creases in one closing; the topper finishes waist, seat and fly.
- **Vertical presses and blade presses** — paired heated bucks close on collars, cuffs, fronts and pleats for a sharp, repeatable set.
- **Steam tunnel** — garments on hangers pass through steam and hot air; high-volume crease removal for knitwear and packed goods, no sharp creasing.
- **Fusing press** (flat-bed or continuous) — belongs to making rather than finishing, but its temperature–pressure–time discipline (Topic 7) is the same physics.

4. Pressing faults

FAULT	CAUSE AND PREVENTION

Shine / glazing	Too much heat + pressure + dryness on wool and synthetics. Prevent: press cloth, steam, lower temperature, suede/needle-point boards.
Scorching	Temperature beyond the fibre's tolerance, or a stationary iron. Prevent: correct setting, keep the iron moving, soleplate cleaning.
Imprinted seams/texture	Seam allowances or pocket shapes pressed through to the face. Prevent: paper strips under allowances, shaped bucks, vacuum before lifting.
Water spots / steam marks	Wet steam, dirty boiler water, pressing before the part is dry-set. Prevent: boiler maintenance, vacuum drying, test on the style's fabric.
Crease in the wrong place	Misalignment on the press — and on thermoplastic fibres the wrong crease heat-sets permanently. Prevent: alignment guides and first-piece checks.

5. Finishing beyond pressing

Finishing is the whole end-of-line: **thread trimming** (loose ends clipped — a defect class of its own in inspection), **stain removal** (spotting with the correct solvent for oil, chalk or rust marks — wrong solvent sets the stain), **final measurement and visual check** against the spec, **repair routing** (faults return to the operation that made them), and attachment of labels, tags and spare buttons per the buyer's manual. Only then does the garment pass to folding and packing.

Glossary

TERM	MEANING
Under-pressing	In-process pressing of parts and seams during manufacture.
Buck	A shaped pressing form (board, cushion or metal shape) supporting the garment part being pressed.
Glazing	Heat-and-pressure shine on the fabric face; a pressing fault on wool and synthetics.
Form finisher	Machine that dresses a garment on an inflatable form and finishes it with internal steam and hot air.

Key points and summary

- Pressing = heat + moisture + pressure + vacuum/cooling, dosed to the fibre's tolerance.
- Under-pressing during making is a quality operation; top pressing before packing is presentation — both are specified.

- The classic faults are shine, scorch, imprint and spots; most are prevented by the press cloth, correct temperature and vacuum discipline.
- Finishing includes trimming, spotting, checking and tagging — the garment is not finished when the pressing ends.

Practice

Answer in your own words; brief model answers follow.

1. A wool jacket front comes off the press shiny. Identify the fault, its three contributing variables, and the corrected method.
2. Why can a wrongly placed crease in a polyester trouser not simply be pressed out and re-creased?
3. Match equipment to task: (a) trouser leg creases for 800 pairs/day, (b) a lined blazer, (c) crease removal from packed T-shirts in bulk.
4. State the four pressing variables in the order a finisher applies them on a vacuum table.

Answer key

1. Glazing (shine): excessive heat, pressure and dryness together. Correct by using steam, a press cloth, lower temperature and vacuum setting — never a hot dry iron pressed hard on wool.
2. Polyester is thermoplastic and heat-sets: the first crease was permanently set into the fibre structure at pressing temperature, so a new crease can be added but the old one resists removal — a permanent quality defect.
3. (a) A legger press (with a topper for the upper part). (b) Form finisher plus hand touch-up on bucks. (c) Steam tunnel.
4. Heat and moisture (steam) to plasticise, pressure to shape, then vacuum/cooling to dry and fix the set before the part is moved.